LUND UNIVERSITY
Mathematics, Faculty of Science

Number theory
MATM15
HT 2019
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Written examination
January 18, 2020, 8:00-13:00

1. Determine, using Legendre symbols, whether the congruences

$$x^2 + 5x + 9 \equiv 0 \pmod{883} \quad \text{and} \quad x^2 + 5x + 9 \equiv 0 \pmod{887},$$

respectively, are solvable or not. (Note that 883 and 887 are primes.)

2. Determine the infinite continued fraction representation of the number $\dfrac{\sqrt{3}-1}{2}$.

3. Prove that

$$\sum_{d|n} \mu(d)\tau(d) = (-1)^{\omega(n)}$$

for any positive integer n, where μ is the Möbius function, τ is the divisor function and $\omega(n)$ is the number of distinct prime divisors of n.

4. Let p be a prime of the form $p = 2^n + 1$ for a positive integer n. Prove that every integer a with $\gcd(a, p) = 1$ is either a quadratic residue or a primitive root (mod p).

5. Let p be a prime and define

$$S_n = \sum_{j=1}^{p} j^n$$

for any positive integer n. Prove that either $S_n \equiv 0 \pmod{p}$ or $S_n \equiv -1 \pmod{p}$. For which n does the second alternative hold?

6. Prove that a prime p can be represented as

$$p = 3x^2 - y^2$$

for $x, y \in \mathbb{Z}$ if and only if either $p \in \{2, 3\}$ or $p \equiv 11 \pmod{12}$.

Give careful motivations for your solutions!